TOPIC 3 · CONSERVATION LAWS

Solved Problems: Official Collection CL1 to CL5

Worked solutions to the first problems of the conservation-laws collection of the Department of Thermal and Fluid Engineering, Universidad Carlos III de Madrid (CL1 to CL5 so far; the rest of the collection is added in batches). Every problem restates the official statement, then follows four phases: (1) hypotheses, degrees of freedom and data; (2) coordinates, frames and changes of variable; (3) step-by-step derivation; (4) interpretation, limits and dimensional check. Discrepancies found in the handwritten official solutions are flagged in the corresponding problem, and anything beyond the course material is marked as an extension. References to Notes.pdf are to Chapter 3 (Conservation laws).

PROBLEM CL1

Airfoil Wake: Streamline Displacement and Drag

Original statement · CL1.pdf

Control volume | Linear momentum | Wake | Drag

OFFICIAL PROBLEM STATEMENT

The experimental tests of an airfoil in a wind tunnel indicate that the velocity profile upstream remains uniform, whereas behind the airfoil there exists a wake region with significant velocity deficit whose thickness increases downstream, resulting in the flow field sketched in the figure of CL1.pdf. For the airfoil tested, the velocity measurements at a given downstream section reveal that the velocity in the wake varies linearly from the value $u = U/2$ at the centerline to reach the value $u = U$ at the edge of the wake, located at $y = y_0$. To analyze the problem, consider a control volume limited laterally by the streamlines that extend from $x \to -\infty$, $y = \pm y_{-\infty}$ ($y_{-\infty} < y_0$) to $x = x_0$, $y = \pm y_0$ and internally by the surface of the airfoil. Assuming that the air behaves as an incompressible fluid with density ρ and that viscous stresses are negligible both in the wake and along the bounding streamlines, obtain the value of $y_{-\infty}$ as a function of y_0, as well as the value of the drag on the airfoil as a function of ρ, y_0 and U. In the analysis, assume that on the outer boundary of the control volume the pressure differences from the ambient value p_∞ are negligible, and so is the effect of gravity on the air flow.

Source: CL1.pdf (page 1). Notes.pdf, Chapter 3: integral mass balance Eqs. 3.10-3.11, integral momentum balance Eqs. 3.35-3.36, force of the fluid on a body Eq. 3.32.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Degrees of Freedom and Data

- Steady flow ($\partial/\partial t = 0$), planar (everything is computed per unit span in the direction z normal to the figure), incompressible (constant ρ), no gravity.
- Viscous stresses are negligible in the wake and on the bounding streamlines (given); pressure differences $p - p_\infty$ are negligible on the whole outer boundary of the control volume (given). Pressure and viscous stresses on the airfoil surface are **not** neglected: they are the drag.
- Wake profile at the section $x = x_0$, symmetric about $y = 0$: $v_x = \frac{U}{2}\left(1 + \frac{|y|}{y_0}\right)$ for $|y| \leq y_0$ and $v_x = U$ for $|y| > y_0$. The only unknowns are the scalar $y_{-\infty}$ and the drag D.

Symbol	Meaning	SI unit
U	uniform upstream velocity	m/s
y_0	half-width of the wake at $x = x_0$	m
$y_{-\infty}$	half-width of the stream tube far upstream	m
ρ	air density	kg/m^3
D	drag per unit span	N/m

Dimensional check of the unknown: $[\rho U^2 y_0] = (\mathrm{kg/m^3})(\mathrm{m^2/s^2})\,\mathrm{m} = \mathrm{kg/s^2} = \mathrm{N/m}$, a force per unit length, as required for a planar problem.

PHASE 2

Coordinates, Frames and Changes of Variable

- Inertial frame fixed to the airfoil and the wind tunnel, Cartesian basis $(\mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z)$ with $\mathbf{e}_x$ along the incoming stream and $\mathbf{e}_z = \mathbf{e}_x \wedge \mathbf{e}_y$. The control volume is fixed in this frame, so $\mathbf{v}_c = 0$ in the universal balances (Eqs. 3.10, 3.35 reduce to Eqs. 3.11, 3.36).
- Boundary of the control volume: Σ_u (section far upstream, $|y| < y_{-\infty}$, $\mathbf{n} = -\mathbf{e}_x$), Σ_d (section $x = x_0$, $|y| < y_0$, $\mathbf{n} = +\mathbf{e}_x$), Σ_l (the two bounding streamlines), and Σ_w (airfoil surface, with $\mathbf{n}$ pointing out of the fluid, i.e. into the airfoil).

- The force of the fluid on the airfoil is $\mathbf{F} = \int_{\Sigma_w} (-p\,\mathbf{n}_w + \bar{\bar{\tau}}' \cdot \mathbf{n}_w)\, d\sigma$ with $\mathbf{n}_w = -\mathbf{n}$ pointing out of the airfoil (Eq. 3.32). The force of the airfoil on the fluid is therefore $-\mathbf{F}$.
- Change of variable used in the integral of the wake: $s = 1 + y/y_0$, $ds = dy/y_0$, so that $y \in [0, y_0]$ maps to $s \in [1, 2]$.

PHASE 3

Step-by-Step Derivation

3.1 Mass balance: the width $y_{-\infty}$ of the incoming stream tube

Why this tool: the control volume is fixed and the flow is steady, so the integral continuity equation (Notes Eq. 3.11) reduces to $\int_{\Sigma_c} \rho\,\mathbf{v}\cdot\mathbf{n}\, d\sigma = 0$. It is the only relation that links the unknown $y_{-\infty}$ to the measured profile.

On Σ_l (streamlines) and on Σ_w (impermeable wall) $\mathbf{v}\cdot\mathbf{n} = 0$, so only Σ_u and Σ_d contribute. On Σ_u, $\mathbf{v}\cdot\mathbf{n} = -U$ over a width $2y_{-\infty}$:

$$\int_{\Sigma_u} \rho\,\mathbf{v}\cdot\mathbf{n}\, d\sigma = -\rho U\,(2y_{-\infty}).$$

On Σ_d, $\mathbf{v}\cdot\mathbf{n} = v_x$ and, by symmetry, the integral is twice the integral over $0 \le y \le y_0$. The antiderivative of $1 + y/y_0$ is $y + y^2/(2y_0)$, and Barrow's rule between $y = 0$ and $y = y_0$ gives:

$$\int_{\Sigma_d} \rho\,\mathbf{v}\cdot\mathbf{n}\, d\sigma = 2\rho \int_0^{y_0} \frac{U}{2}\left(1 + \frac{y}{y_0}\right) dy = \rho U \left[y + \frac{y^2}{2y_0}\right]_0^{y_0} = \rho U \left(y_0 + \frac{y_0}{2}\right) = \frac{3}{2}\rho U y_0.$$

Adding both contributions and setting the sum to zero:

$$-2\rho U y_{-\infty} + \frac{3}{2}\rho U y_0 = 0 \implies \boxed{y_{-\infty} = \frac{3}{4}\,y_0.}$$

3.2 Linear momentum balance: the drag

Why this tool: the force of the fluid on a body is not accessible from the (unknown) pressure and shear on its surface, but it appears in the integral momentum balance (Notes Eq. 3.36, steady, fixed control volume, no gravity) as the only unknown surface term:

$$\int_{\Sigma_c} \rho\, \mathbf{v}\, (\mathbf{v}\cdot\mathbf{n})\, d\sigma = \int_{\Sigma_c} \left(-p\,\mathbf{n} + \bar{\bar{\tau}}'\cdot\mathbf{n}\right) d\sigma .$$

Because Σ_c is a closed surface, $\int_{\Sigma_c} p_\infty \mathbf{n}\, d\sigma = 0$, and p may be replaced by $p - p_\infty$ on the right-hand side. On Σ_u, Σ_d and Σ_l the difference $p - p_\infty$ and the viscous stresses are negligible (hypothesis), so the only surviving term is the one on Σ_w, which equals the force of the airfoil on the fluid, $-\mathbf{F}$. The x component gives the drag $D = F_x$:

$$D = -\int_{\Sigma_u+\Sigma_d+\Sigma_l} \rho\, v_x\, (\mathbf{v}\cdot\mathbf{n})\, d\sigma .$$

On Σ_l, $\mathbf{v}\cdot\mathbf{n} = 0$. On Σ_u, $v_x = U$ and $\mathbf{v}\cdot\mathbf{n} = -U$, hence with 3.1:

$$\int_{\Sigma_u} \rho\, v_x (\mathbf{v}\cdot\mathbf{n})\, d\sigma = -\rho U^2\,(2y_{-\infty}) = -\frac{3}{2}\rho U^2 y_0 .$$

On Σ_d, with the substitution $s = 1 + y/y_0$, $dy = y_0\, ds$, and Barrow's rule between $s = 1$ and $s = 2$:

$$\int_{\Sigma_d} \rho\, v_x^2\, d\sigma = 2\rho\,\frac{U^2}{4}\int_0^{y_0}\left(1+\frac{y}{y_0}\right)^2 dy = \frac{\rho U^2}{2}\,y_0\int_1^2 s^2\, ds = \frac{\rho U^2 y_0}{2}\left[\frac{s^3}{3}\right]_1^2 = \frac{\rho U^2 y_0}{2}\cdot\frac{8-1}{3} = \frac{7}{6}\rho U^2 y_0 .$$

Therefore:

$$D = -\left(-\frac{3}{2}+\frac{7}{6}\right)\rho U^2 y_0 = -\left(-\frac{9}{6}+\frac{7}{6}\right)\rho U^2 y_0 \;\Longrightarrow\; \boxed{D = \frac{1}{3}\,\rho\, U^2\, y_0} \quad \text{per unit span, in the direction } +\mathbf{e}_x .$$

For a wake symmetric about $y = 0$ (a symmetric airfoil at zero angle of attack, which the statement implies by giving a symmetric profile) the y component of the same balance gives $F_y = 0$: there is no lift.

PHASE 4

Interpretation, Limits and Dimensional Check

- Physical picture: the wake carries less mass flux ($\frac{3}{2}\rho U y_0$) than the undisturbed stream would through the same width ($2\rho U y_0$); the missing flux is the volume displaced sideways by the wake, which is why the bounding streamlines are at $y_{-\infty} = \frac{3}{4} y_0 < y_0$ far upstream. The momentum deficit left in the wake is the drag.
- Limit $y_0 \to 0$: no wake, $y_{-\infty} \to 0$ and $D \to 0$, consistent with d'Alembert's paradox for a frictionless attached flow.
- Dimensions (SI): $[y_{-\infty}] = [y_0] = \mathrm{m}$; $[D] = [\rho U^2 y_0] = \mathrm{N/m}$.
- Numerical cross-check (midpoint quadrature with 2×10^5 cells of the defining integrals, $U = 1$ m/s, $\rho = 1.2$ kg/m^3, $y_0 = 0.35$ m): $y_{-\infty}/y_0 = 0.75000$ and $D = 0.140000$ N/m, equal to $\rho U^2 y_0 / 3 = 0.14$ N/m.
- Agreement with the official solution: CL1.pdf obtains $y_{-\infty} = \frac{3}{4} y_0$ and $\mathbf{F}_{\text{airfoil}} = \frac{1}{3}\rho U^2 y_0 \, \mathbf{e}_x$, the same results.

Extension (beyond the official solution)

With the momentum thickness $\theta = \int_{-y_0}^{y_0} \frac{v_x}{U}\left(1 - \frac{v_x}{U}\right) dy = 2y_0 \int_0^1 \frac{1-\sigma^2}{4}\, d\sigma = \frac{y_0}{3}$ (substituting $\sigma = y/y_0$), the result reads $D = \rho U^2 \theta$: the classical momentum-deficit formula for the drag from a wake survey.

PROBLEM CL2

Bore over a Stagnant Water Layer

Original statement · CL2.pdf

Moving frame | Mass and momentum | Hydrostatic pressure | Froude number

OFFICIAL PROBLEM STATEMENT

The bore sketched in the figure of CL2.pdf moves under the action of gravity with constant velocity U over an initially stagnant water layer of thickness h_0, leaving behind a layer of thickness h_1 moving with velocity U_1. Assuming that in the air both viscous stresses and pressure variations with respect to the ambient value p_a are negligible, and so are viscous stresses on the bottom wall, find the values of U_1 and h_1 as a function of U, h_0 and g. As a preliminary computation, obtain the pressure distributions both upstream and downstream from the bore, verifying that they can be expressed in the form $p = p_a + \rho g(h_0 - z)$ and $p = p_a + \rho g(h_1 - z)$, respectively, where z represents the vertical distance to the bottom wall.

Source: CL2.pdf (page 1). Notes.pdf, Chapter 3: integral mass balance Eqs. 3.10-3.11, integral momentum balance Eqs. 3.35-3.36.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Degrees of Freedom and Data

- Planar flow (quantities per unit width normal to the figure) of an incompressible liquid of density ρ, under gravity $\mathbf{g} = -g\,\mathbf{e}_z$. The bore advances towards $-\mathbf{e}_x$ (the x axis points to the right in the figure) with constant velocity U over the stagnant layer of thickness h_0; behind it the layer has thickness h_1 and uniform velocity $-U_1\mathbf{e}_x$.
- Far from the front, in each region, the velocity is horizontal and uniform and the free surface is horizontal at pressure p_a. Viscous stresses are negligible (given); the ratio of viscous to inertial forces is of order $\mu/(\rho U h) \ll 1$ for a water layer.
- The two unknowns are h_1/h_0 and U_1/U; the data are U, h_0, g and ρ.

Symbol	Meaning	SI unit
U	velocity of the bore (relative to the stagnant layer)	m/s

Symbol	Meaning	SI unit
U_1	velocity of the water behind the bore	m/s
h_0, h_1	layer thickness ahead of and behind the bore	m
p_a	ambient pressure acting on the free surface	Pa
g	gravitational acceleration	m/s^2
$F = U^2/(gh_0)$	Froude number of the bore (dimensionless)	1

Dimensional check: $[U^2/(gh_0)] = (\mathrm{m}^2/\mathrm{s}^2)/\big((\mathrm{m}/\mathrm{s}^2)\,\mathrm{m}\big) = 1$.

PHASE 2

Coordinates, Frames and Changes of Variable

- Laboratory frame L: Cartesian basis $(\mathbf{e}_x, \mathbf{e}_z)$ with z measured from the bottom wall. The flow is unsteady in L because the front moves.
- Frame of the bore B: it moves with the velocity $-U\,\mathbf{e}_x$ relative to L. Since this velocity is constant, B is inertial and no inertial forces appear in the balances. Velocities in B are $\mathbf{v}_B = \mathbf{v}_L + U\,\mathbf{e}_x$: the liquid ahead has $\mathbf{v}_B = U\,\mathbf{e}_x$ and the liquid behind has $\mathbf{v}_B = (U - U_1)\,\mathbf{e}_x$. In B the front is stationary, so the flow is steady.
- Control volume fixed in B (so $\mathbf{v}_c = 0$; Eqs. 3.11 and 3.36): Σ_0 (vertical section ahead of the front, $\mathbf{n} = -\mathbf{e}_x$, area h_0 per unit width), Σ_1 (vertical section behind, $\mathbf{n} = +\mathbf{e}_x$, area h_1), Σ_b (bottom wall, $\mathbf{n} = -\mathbf{e}_z$) and Σ_s (free surface, material and steady in B). On Σ_b and Σ_s, $\mathbf{v}\cdot\mathbf{n} = 0$.

PHASE 3

Step-by-Step Derivation

3.1 Preliminary result: hydrostatic pressure upstream and downstream

Why this tool: in the uniform regions the velocity is horizontal, so the z component of the integral momentum balance (Notes Eq. 3.36) has no momentum flux and reduces to a force balance. Take a fixed control volume formed by the liquid column of width Δx (per unit width) between a height z and the free surface $z = h_0$ (upstream region). Its faces are: the bottom face at height z ($\mathbf{n} = -\mathbf{e}_z$), the free surface ($\mathbf{n} = +\mathbf{e}_z$, $p = p_a$) and two vertical faces ($\mathbf{n} = \mp\mathbf{e}_x$).

The z component of $\rho\,\mathbf{v}\,(\mathbf{v}\cdot\mathbf{n})$ vanishes on every face because $v_z = 0$. The pressure force has z component $+p(z)\Delta x$ on the bottom face and $-p_a\Delta x$ on the free surface; the pressure on the vertical faces points along $\pm\mathbf{e}_x$ and viscous stresses are negligible. The

weight of the column is $-\rho g\,(h_0 - z)\,\Delta x$. Hence

$$0 = p(z)\,\Delta x - p_a\,\Delta x - \rho g\,(h_0 - z)\,\Delta x \implies \boxed{p = p_a + \rho g\,(h_0 - z)} \quad \text{(upstream)}.$$

The same argument with the surface at $z = h_1$ gives $p = p_a + \rho g\,(h_1 - z)$ downstream.

3.2 Mass balance in the frame of the bore

Why this tool: steady flow and fixed control volume in B, so $\int_{\Sigma_c} \rho\,\mathbf{v}\cdot\mathbf{n}\,d\sigma = 0$ (Notes Eq. 3.11). Only Σ_0 and Σ_1 are crossed by the flow:

$$-\rho\,U\,h_0 + \rho\,(U - U_1)\,h_1 = 0 \implies U\,h_0 = (U - U_1)\,h_1. \qquad \text{(c1)}$$

3.3 Momentum balance in the frame of the bore

Why this tool: the unknown h_1 appears through the pressure forces, which only the momentum balance (Notes Eq. 3.36) contains. Since $\int_{\Sigma_c} p_a\,\mathbf{n}\,d\sigma = 0$, replace p by $p - p_a$. The x component of the left-hand side is

$$\int_{\Sigma_c} \rho\,v_x\,(\mathbf{v}\cdot\mathbf{n})\,d\sigma = \rho U\,(-U h_0) + \rho\,(U - U_1)\,(U - U_1)\,h_1 = -\rho U^2 h_0 + \rho\,(U - U_1)^2\,h_1.$$

Right-hand side: on Σ_s, $p - p_a = 0$; on Σ_b the normal is $-\mathbf{e}_z$, with no x component; the viscous stresses are negligible (hypothesis); gravity has no x component. The pressure on Σ_0 and Σ_1 is the hydrostatic distribution of 3.1, with $\mathbf{n} = -\mathbf{e}_x$ and $+\mathbf{e}_x$ respectively. The antiderivative of $\rho g\,(h - z)$ is $\rho g\,(hz - z^2/2)$ and Barrow's rule between $z = 0$ and $z = h$ gives $\rho g\,(h^2 - h^2/2) = \rho g h^2/2$:

$$-\int_{\Sigma_0} (p - p_a)\,\mathbf{n}\,d\sigma\cdot\mathbf{e}_x = +\int_0^{h_0} \rho g\,(h_0 - z)\,dz = \frac{\rho g h_0^2}{2}, \qquad -\int_{\Sigma_1} (p - p_a)\,\mathbf{n}\,d\sigma\cdot\mathbf{e}_x = -\int_0^{h_1} \rho g\,(h_1 - z)\,dz = -\frac{\rho g h_1^2}{2}.$$

$$-\rho U^2 h_0 + \rho\,(U-U_1)^2\,h_1 = \frac{\rho g}{2}\left(h_0^2 - h_1^2\right). \qquad \text{(c2)}$$

3.4 Solution for h_1/h_0 and U_1/U

From (c1), $U - U_1 = U h_0/h_1$, so $(U-U_1)^2\,h_1 = U^2 h_0^2/h_1$. Substituting into (c2) and dividing by ρ:

$$U^2 h_0\left(\frac{h_0}{h_1} - 1\right) = \frac{g}{2}\,(h_0 - h_1)(h_0 + h_1) \;\Longrightarrow\; U^2 h_0\,\frac{h_0 - h_1}{h_1} = \frac{g}{2}\,(h_0 - h_1)(h_0 + h_1).$$

For a genuine bore $h_1 \neq h_0$ (the case $h_1 = h_0$ is the undisturbed layer), so the factor $h_0 - h_1$ can be cancelled. With $r \equiv h_1/h_0$ and $F = U^2/(g h_0)$, dividing by $g h_0^2$ gives $F/r = (1+r)/2$, that is

$$r^2 + r - 2F = 0 \;\Longrightarrow\; \boxed{\frac{h_1}{h_0} = \frac{1}{2}\left(\sqrt{1+8F} - 1\right)} \qquad \text{(the negative root is rejected since } h_1 > 0\text{)}.$$

The velocity behind follows from (c1): $U_1/U = 1 - h_0/h_1 = 1 - 1/r = (r-1)/r$. With $s \equiv \sqrt{1+8F}$, $r = (s-1)/2$ and

$$\boxed{\frac{U_1}{U} = 1 - \frac{2}{s-1} = \frac{\sqrt{1+8F} - 3}{\sqrt{1+8F} - 1}}, \qquad F = \frac{U^2}{g h_0}.$$

Erratum in the official solution (reading of the handwriting)

CL2.pdf gives $h_1/h_0 = \frac{1}{2}(\sqrt{1+8F}-1)$, which agrees with the result above. For the velocity it gives, as legible in the scan, $U_1/U = \frac{3}{2}\left(\frac{1}{3}\sqrt{1+8F}-1\right) = (\sqrt{1+8F}-3)/2 = r-1$. This contradicts the official continuity equation $Uh_0 = (U-U_1)h_1$, which requires $U_1/U = (r-1)/r$; the two expressions coincide only for $r=1$. The check that proves it: $(U-U_1)h_1 = Uh_0 > 0$ forces $U_1 < U$, whereas $r-1>1$ for $F>3$ (for $F=4$, $r-1 = 1.3723$ against the correct $U_1/U = 0.5785$, and the mass residual of (c1) with the scan formula is $Uh_0-(U-U_1)h_1 \neq 0$). The computed value is kept.

PHASE 4

Interpretation, Limits and Dimensional Check

- The bore exists only for $F>1$: $r>1 \iff s>3 \iff F>1$. F is the Froude number of the front relative to the shallow-water wave speed $\sqrt{gh_0}$; a front slower than $\sqrt{gh_0}$ cannot raise the level.
- Weak bore, $F = 1+\epsilon$: $r \approx 1+\frac{2}{3}\epsilon$ and $U_1/U \approx \frac{2}{3}\epsilon \to 0$; the front degenerates into an infinitesimal gravity wave travelling at $\sqrt{gh_0}$.
- Strong bore, $F \gg 1$: $r \approx \sqrt{2F}$, so $h_1 \approx h_0\sqrt{2}\,U/\sqrt{gh_0}$ and $U_1 \to U$: the liquid behind moves almost with the bore.
- Dimensions (SI): both sides of (c2) are forces per unit width, $[\rho U^2 h_0] = [\rho g h^2] = \mathrm{N/m}$; F, r and U_1/U are dimensionless.
- Numerical cross-check: $F=4$, $h_0 = 0.5$ m, $g = 9.81$ m/s^2 give $U = 4.4294$ m/s, $h_1/h_0 = 2.3723$ ($h_1 = 1.1861$ m), $U_1/U = 0.57846$ ($U_1 = 2.5623$ m/s); the residuals of (c1) and (c2) are below 10^{-14} in SI units.

Extension (beyond the official solution)

Mechanical energy is not conserved across the bore: Bernoulli's equation between both sections would give a different relation. The head lost is the classical hydraulic-jump value $\Delta H = (h_1-h_0)^3/(4h_0h_1)$, always positive for $h_1 > h_0$.

PROBLEM CL3

Double-Elbow Mouthpiece: Exit Velocity, Force and Moment

Original statement · CL3.pdf

Control volume | Linear momentum | Angular momentum | Gauge pressure

> **OFFICIAL PROBLEM STATEMENT**
>
> The double-elbow mouth piece of exit cross section A_o shown in the figure of CL3.pdf is to be installed at the end of a circular pipe of cross section A_p to redirect the outflow of water into the open atmosphere. Once the mouth piece is installed, the uniform values U_p and p_p are measured for the flow velocity and the pressure at the end of the pipe (i.e., just at the entrance of the mouth piece). Find the value of the uniform velocity at the exit section, U_o, as well as the force exerted on the mouth piece and its corresponding moment with respect to the entrance section. (Handwritten addition: neglect the effect of gravity.)
>
> Source: CL3.pdf (page 1). Notes.pdf, Chapter 3: integral mass balance Eq. 3.11, integral momentum balance Eq. 3.36, integral angular momentum balance Eqs. 3.37-3.38, force and moment on a body Eqs. 3.32-3.33.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Degrees of Freedom and Data

- Steady flow of an incompressible liquid (water, constant ρ), uniform velocity and pressure over the entrance section (area A_p) and over the exit section (area A_o); no gravity (handwritten addition to the statement).
- The water discharges into the atmosphere, so the pressure over the exit section is the ambient pressure p_a. Viscous stresses over the entrance and exit sections are negligible (uniform velocity, no gradients).
- The wall of the mouth piece is thin and its outer face is in contact with the atmosphere at p_a. The 'force exerted on the mouth piece' is the net force of the water inside plus the atmosphere outside, which is how the official solution treats it (it adds the term in p_a explicitly). If the atmosphere were ignored, the force would change by $p_a\,(A_p - A_o)\,\mathbf{e}_x$.

- Geometry from the figure: the exit axis is displaced a vertical distance H below the axis of the entrance, and both the entrance flow and the exit flow are along $+\mathbf{e}_x$ (the mouth piece is S-shaped).

Symbol	Meaning	SI unit
A_p, A_o	entrance and exit areas	m^2
U_p, p_p	uniform velocity and pressure at the entrance	m/s, Pa
U_o, p_a	exit velocity (unknown) and ambient pressure	m/s, Pa
H	vertical offset between the entrance and exit axes	m
$\mathbf{F}_{MP}$, $\mathbf{M}_{MP}$	force and moment on the mouth piece (unknowns)	N, N·m

PHASE 2

Coordinates, Frames and Changes of Variable

- Inertial frame fixed to the pipe, Cartesian basis $(\mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z)$: $\mathbf{e}_x$ along the pipe axis in the flow direction, $\mathbf{e}_y$ vertical upwards and $\mathbf{e}_z = \mathbf{e}_x \wedge \mathbf{e}_y$ out of the plane of the figure. The origin O is the centre (centroid) of the entrance section; moments are taken about O.
- The centre of the exit section is at the position $\mathbf{x}_o = X\,\mathbf{e}_x - H\,\mathbf{e}_y$ relative to O; X is the horizontal offset, which will drop out. The position of a point of a circular section relative to its centre is denoted $\boldsymbol{\eta}$, with $\int \boldsymbol{\eta}\, d\sigma = 0$.
- Fixed control volume: the water inside the mouth piece, with boundary Σ_p (entrance, $\mathbf{n} = -\mathbf{e}_x$), Σ_o (exit, $\mathbf{n} = +\mathbf{e}_x$) and Σ_w (inner wall, $\mathbf{n}$ out of the fluid). Since $\mathbf{v}_c = 0$, Eqs. 3.11, 3.36 and 3.38 apply with $\mathbf{v}_c = 0$.
- Since $\int_{\Sigma_c} p_a\,\mathbf{n}\, d\sigma = 0$ on the closed surface, p is replaced by $p - p_a$ in the balances. The force of the water and the atmosphere on the (thin) wall is then $\mathbf{F}_{MP} = \int_{\Sigma_w} \left[(p - p_a)\,\mathbf{n} - \bar{\bar{\tau}}' \cdot \mathbf{n}\right] d\sigma$ and the force of the wall on the water is $-\mathbf{F}_{MP}$.

PHASE 3

Step-by-Step Derivation

3.1 Exit velocity from mass conservation

Why this tool: the velocities at both sections are uniform and the flow is steady, so the continuity balance (Notes Eq. 3.11) relates them directly. On Σ_w, $\mathbf{v} \cdot \mathbf{n} = 0$:

$$-\rho\, U_p A_p + \rho\, U_o A_o = 0 \implies \boxed{U_o = \frac{A_p}{A_o}\, U_p}.$$

3.2 Force on the mouth piece: linear momentum

Why this tool: the force on the wall is an unknown surface integral of the (non-uniform) wall pressure; the momentum balance (Notes Eq. 3.36) replaces it by quantities known at the two sections. Left-hand side: the momentum flux is only through Σ_p and Σ_o, and both velocities are along $\mathbf{e}_x$:

$$\int_{\Sigma_c} \rho\, \mathbf{v}\, (\mathbf{v}\cdot\mathbf{n})\, d\sigma = \left(-\rho U_p^2 A_p + \rho U_o^2 A_o\right) \mathbf{e}_x = \rho U_p A_p\, (U_o - U_p)\, \mathbf{e}_x,$$

where $\rho U_o^2 A_o = (\rho U_o A_o)\, U_o = \rho U_p A_p\, U_o$ by 3.1. Right-hand side: on Σ_p the force is $-(p_p - p_a)(-\mathbf{e}_x) A_p = +(p_p - p_a) A_p\, \mathbf{e}_x$; on Σ_o it vanishes because $p - p_a = 0$; viscous stresses are negligible on both sections; on Σ_w it is $-\mathbf{F}_{MP}$. Hence

$$\rho U_p A_p\, (U_o - U_p)\, \mathbf{e}_x = (p_p - p_a)\, A_p\, \mathbf{e}_x - \mathbf{F}_{MP} \implies \boxed{\mathbf{F}_{MP} = \left[(p_p - p_a)\, A_p + \rho\, U_p\, (U_p - U_o)\, A_p\right] \mathbf{e}_x.}$$

The y and z components vanish: no momentum flux and no pressure force on the sections has a component along $\mathbf{e}_y$ or $\mathbf{e}_z$, so the balance gives $F_{MP,y} = F_{MP,z} = 0$.

3.3 Moment on the mouth piece: angular momentum about O

Why this tool: the moment of the wall forces is also unknown; the angular momentum balance (Notes Eq. 3.38, fixed control volume, no gravity) gives it from the fluxes through the sections:

$$\int_{\Sigma_c} \rho\, \left[(\mathbf{x} - \mathbf{x}_0) \wedge \mathbf{v}\right] (\mathbf{v}\cdot\mathbf{n})\, d\sigma = \sum \mathbf{M}_{O,\text{ext}}, \qquad \mathbf{x}_0 = \mathbf{x}_O.$$

Entrance Σ_p: $\mathbf{x} - \mathbf{x}_O = \boldsymbol{\eta}$, $\mathbf{v} = U_p \mathbf{e}_x$ uniform and $\int \boldsymbol{\eta}\, d\sigma = 0$, so both the angular momentum flux $-\rho U_p^2 \left(\int \boldsymbol{\eta}\, d\sigma\right) \wedge \mathbf{e}_x$ and the moment of the pressure force $(p_p - p_a) \left(\int \boldsymbol{\eta}\, d\sigma\right) \wedge \mathbf{e}_x$ vanish. Exit Σ_o: $\mathbf{x} - \mathbf{x}_O = \mathbf{x}_o + \boldsymbol{\eta}$ and $\int \boldsymbol{\eta}\, d\sigma = 0$, so with $\mathbf{x}_o \wedge \mathbf{e}_x =$

$(X\,\mathbf{e}_x - H\,\mathbf{e}_y) \wedge \mathbf{e}_x = -H\,\mathbf{e}_y \wedge \mathbf{e}_x = +H\,\mathbf{e}_z$ (since $\mathbf{e}_y \wedge \mathbf{e}_x = -\mathbf{e}_z$), the flux is

$$\int_{\Sigma_o} \rho\,[(\mathbf{x}_o + \boldsymbol{\eta}) \wedge U_o \mathbf{e}_x]\,U_o\,d\sigma = \rho\,U_o^2\,A_o\,(\mathbf{x}_o \wedge \mathbf{e}_x) = \rho\,U_o^2\,H\,A_o\,\mathbf{e}_z .$$

The pressure moment on Σ_o vanishes ($p - p_a = 0$), viscous stresses are negligible on the sections, and the moment of the wall force on the water is $-\mathbf{M}_{MP}$. Hence $\rho U_o^2 H A_o\,\mathbf{e}_z = -\mathbf{M}_{MP}$ and

$$\boxed{\mathbf{M}_{MP} = -\rho\,U_o^2\,H\,A_o\,\mathbf{e}_z = -\rho\,\frac{A_p^2}{A_o}\,U_p^2\,H\,\mathbf{e}_z} \quad \text{about the centre of the entrance section.}$$

PHASE 4

Interpretation, Limits and Dimensional Check

- $A_o < A_p$ implies $U_o > U_p$: the nozzle accelerates the liquid, and the momentum flux $\rho U_p A_p\,(U_o - U_p) > 0$ is the reaction that the wall supplies on the water.
- The moment is non-zero only because the exit axis is displaced by H from the entrance axis: for $H \to 0$ the jet leaves along the line of the entrance axis and $\mathbf{M}_{MP} \to 0$. The sign is negative: with the exit below the entrance, the moment is clockwise as seen in the figure (with $\mathbf{e}_z$ pointing towards the reader).
- Limit $A_o \to A_p$: $U_o = U_p$ and $\mathbf{F}_{MP} \to (p_p - p_a) A_p\,\mathbf{e}_x$, the pure pressure force; the moment becomes $-\rho U_p^2 H A_p\,\mathbf{e}_z$.
- Dimensions (SI): $[\rho U_p A_p\,(U_p - U_o)] = (\mathrm{kg/m^3})(\mathrm{m/s})\,\mathrm{m^2}\,(\mathrm{m/s}) = \mathrm{N}$; $[\rho U_o^2 H A_o] = (\mathrm{kg/m^3})(\mathrm{m^2/s^2})\,\mathrm{m}\,\mathrm{m^2} = \mathrm{N \cdot m}$.
- Numerical illustration (hypothetical data): $\rho = 1000$ kg/m^3, $A_p = 0.02$ m^2, $A_o = 0.008$ m^2, $U_p = 3$ m/s, $p_p = 2.5 \times 10^5$ Pa, $p_a = 101325$ Pa, $H = 1.5$ m give $U_o = 7.5$ m/s, $F_{MP,x} = 2703.5$ N and $M_{MP,z} = -675$ N·m.
- Agreement with the official solution: CL3.pdf obtains $U_o = (A_p/A_o)\,U_p$, $\mathbf{F}_{MP} = [(p_p - p_a)A_p + \rho U_p (U_p - U_o) A_p]\,\mathbf{e}_x$ and $\mathbf{M}_{MP} = -\rho U_o^2 H A_o\,\mathbf{e}_z$, the same results.

Extension (beyond the official solution)

If the flow inside were lossless (not stated in the problem), Bernoulli's equation between both sections, $p_p - p_a = \frac{1}{2}\rho\left(U_o^2 - U_p^2\right)$, would give $F_{MP,x} = \frac{1}{2}\rho A_p \left(U_o - U_p\right)^2 \geq 0$. A real mouth piece has losses, and then the measured p_p is the quantity to use.

PROBLEM CL4

Cascade of Aerofoils: Downstream State

Original statement · CL4.pdf

Periodicity | Mass, momentum, energy | Perfect gas | Cascade

OFFICIAL PROBLEM STATEMENT

A planar uniform stream of air with pressure p_1, density ρ_1 and velocity $\mathbf{v}_1 = v_{1x}\mathbf{e}_x$ flows past an array of aerofoils with vertical separation L, exerting on each one a force (per unit length) given by $\mathbf{F} = F_x\mathbf{e}_x + F_y\mathbf{e}_y$. Assuming that the aerofoils are thermally insulated and neglecting the effect of gravity, write the equations that determine the values of the pressure p_2, density ρ_2 and velocity $\mathbf{v}_2 = v_{2x}\mathbf{e}_x + v_{2y}\mathbf{e}_y$ for the uniform stream found sufficiently far downstream.

Source: CL4.pdf (page 1). Notes.pdf, Chapter 3: integral mass balance Eq. 3.11, integral momentum balance Eq. 3.36, integral total energy balance Eq. 3.46, force on a body Eq. 3.32.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Degrees of Freedom and Data

- Steady planar flow of air through an infinite array of identical aerofoils, equally spaced by L in the y direction; the flow is periodic with period L. Far upstream (section 1) and far downstream (section 2) the flow is uniform.
- No gravity; the aerofoils are thermally insulated (adiabatic); viscous stresses and heat conduction are negligible in the uniform sections 1 and 2.
- Perfect gas with specific heat ratio γ: the specific enthalpy is $h = e + p/\rho = \dfrac{\gamma}{\gamma-1}\dfrac{p}{\rho}$. This is the assumption used in the official solution; the statement does not give γ.
- $\mathbf{F}$ is the force that the air exerts on each aerofoil, per unit length normal to the figure, so $[F_x/L]$ is a pressure. There are four unknowns, p_2, ρ_2, v_{2x}, v_{2y}, and therefore four equations are needed: mass, two components of momentum and energy.

Symbol	Meaning	SI unit
p_1, ρ_1, v_{1x}	upstream state (data)	Pa, kg/m^3, m/s

Symbol	Meaning	SI unit
p_2, ρ_2, v_{2x}, v_{2y}	downstream state (unknowns)	Pa, kg/m^3, m/s
L	vertical separation (pitch) of the aerofoils	m
F_x, F_y	force on each aerofoil per unit length	N/m
γ	ratio of specific heats	1

PHASE 2

Coordinates, Frames and Changes of Variable

- Inertial frame fixed to the array, Cartesian basis $(\mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z)$, $\mathbf{e}_x$ along the incoming stream and $\mathbf{e}_y$ along the array.
- Fixed control volume (one period): Σ_1 (section far upstream, height L, $\mathbf{n} = -\mathbf{e}_x$), Σ_2 (section far downstream, height L, $\mathbf{n} = +\mathbf{e}_x$), the two stream surfaces Σ_3 (top) and Σ_4 (bottom), and the surface of the aerofoil Σ_5. $\mathbf{v}_c = 0$, so Eqs. 3.11, 3.36 and 3.46 apply with $\mathbf{v}_c = 0$.
- Periodicity: Σ_4 is Σ_3 translated by $-L\,\mathbf{e}_y$, and the outward normals are opposite at corresponding points ($\mathbf{n}_4 = -\mathbf{n}_3$). Therefore the integral over $\Sigma_3 \cup \Sigma_4$ of any periodic field times the normal vanishes: pressure force, viscous force, heat flux and work of viscous stresses.
- On the stream surfaces $\mathbf{v} \cdot \mathbf{n} = 0$ and on the aerofoil $\mathbf{v} = 0$ (no slip) with $\mathbf{q} \cdot \mathbf{n} = 0$ (insulated). The force of the aerofoil on the fluid is $-\mathbf{F}$ per unit length.

PHASE 3

Step-by-Step Derivation

3.1 Mass

Why this tool: Notes Eq. 3.11 with $\partial/\partial t = 0$. Only Σ_1 and Σ_2 are crossed by the flow, each of height L per unit length:

$$-\rho_1 v_{1x} L + \rho_2 v_{2x} L = 0 \implies \boxed{\rho_1\, v_{1x} = \rho_2\, v_{2x}}. \tag{E1}$$

Denote by $\dot{m}' \equiv \rho_1 v_{1x} L$ the mass flow per unit length through one period.

3.2 Linear momentum, components x and y

Why this tool: the force $\mathbf{F}$ on the aerofoil enters only through the momentum balance (Notes Eq. 3.36). Left-hand side, using (E1) and $v_{1y} = 0$:

$$\int_{\Sigma_c} \rho\, \mathbf{v}\, (\mathbf{v}\cdot\mathbf{n})\, d\sigma = \left(-\rho_1 v_{1x}^2 L + \rho_2 v_{2x}^2 L\right) \mathbf{e}_x + \rho_2 v_{2y} v_{2x} L\, \mathbf{e}_y = \dot{m}' \left[(v_{2x} - v_{1x})\, \mathbf{e}_x + v_{2y}\, \mathbf{e}_y\right].$$

Right-hand side: on Σ_1, $-p_1(-\mathbf{e}_x)L = p_1 L\, \mathbf{e}_x$; on Σ_2, $-p_2 L\, \mathbf{e}_x$; the contributions of Σ_3 and Σ_4 cancel by periodicity; viscous stresses are negligible on Σ_1 and Σ_2; on Σ_5 the force on the fluid is $-\mathbf{F}$. Component by component:

$$\dot{m}'\,(v_{2x} - v_{1x}) = (p_1 - p_2)\, L - F_x \implies \boxed{\frac{F_x}{L} = p_1 - p_2 + \rho_1 v_{1x}\,(v_{1x} - v_{2x})} \qquad \text{(E2)}$$

$$\dot{m}'\, v_{2y} = -F_y \implies \boxed{\frac{F_y}{L} = -\rho_1\, v_{1x}\, v_{2y}} \qquad \text{(E3)}$$

3.3 Total energy

Why this tool: p_2 and ρ_2 are linked by the thermal state of the gas, which only the energy balance (Notes Eq. 3.46) provides. Steady, no gravity, no volumetric heat: the convective flux of $e + v^2/2$ plus the pressure work $p\,\mathbf{v}\cdot\mathbf{n}$ combine into the flux of $\rho\left(h + v^2/2\right)$.

On Σ_3 and Σ_4: $\mathbf{v}\cdot\mathbf{n} = 0$, and conduction and viscous work cancel by periodicity. On Σ_5: $\mathbf{v} = 0$ (no work) and $\mathbf{q}\cdot\mathbf{n} = 0$. On Σ_1 and Σ_2: viscous work and conduction are negligible. What remains is

$$\int_{\Sigma_1+\Sigma_2} \rho \left(h + \frac{v^2}{2}\right) (\mathbf{v}\cdot\mathbf{n})\, d\sigma = 0 \implies \dot{m}' \left[\left(h_2 + \frac{v_{2x}^2 + v_{2y}^2}{2}\right) - \left(h_1 + \frac{v_{1x}^2}{2}\right)\right] = 0,$$

$$\boxed{\frac{\gamma}{\gamma-1}\frac{p_2}{\rho_2}+\frac{v_{2x}^2+v_{2y}^2}{2}=\frac{\gamma}{\gamma-1}\frac{p_1}{\rho_1}+\frac{v_{1x}^2}{2}} \qquad \text{(E4)}$$

Equations (E1) to (E4) are four equations for the four unknowns p_2, ρ_2, v_{2x} and v_{2y}, which answers the question of the statement.

PHASE 4

Interpretation, Limits and Dimensional Check

- Equation (E3) says that the transverse force on an aerofoil (lift) equals minus the rate of change of y momentum of the stream: the flow is deflected by v_{2y} in the direction opposite to the force on the body. Equation (E2) splits the streamwise force into a pressure drop across the array and a momentum change.
- Trivial limit $\mathbf{F}=0$: (E1) to (E4) are satisfied by $\rho_2=\rho_1, p_2=p_1, \mathbf{v}_2=\mathbf{v}_1$ (no body, no change).
- Incompressible limit ($\rho_2=\rho_1$): (E1) gives $v_{2x}=v_{1x}$ and (E2) reduces to $p_1-p_2=F_x/L$.
- Dimensions (SI): $[F_x/L]=(\mathrm{N/m})/\mathrm{m}=\mathrm{Pa}$ and $[\rho_1 v_{1x}^2]=\mathrm{Pa}$; $[\rho_1 v_{1x}\,v_{2y}\,L]=(\mathrm{kg/m^3})(\mathrm{m/s})(\mathrm{m/s})\,\mathrm{m}=\mathrm{N/m}$; $[\gamma p/((\gamma-1)\rho)]=\mathrm{m^2/s^2}=[v^2]$.

Extension (beyond the official solution): closed solution of the four equations

With $m\equiv\rho_1 v_{1x}$, (E1) gives $\rho_2=m/v_{2x}$, (E3) gives $v_{2y}=-F_y/(mL)$ and (E2) gives $p_2=P-m\,v_{2x}$ with $P\equiv p_1+m\,v_{1x}-F_x/L$. Substituting into (E4), with $E\equiv\frac{\gamma}{\gamma-1}\frac{p_1}{\rho_1}+\frac{v_{1x}^2}{2}$, and using that the coefficient of v_{2x}^2 is $\frac{1}{2}-\frac{\gamma}{\gamma-1}=-\frac{\gamma+1}{2(\gamma-1)}$, one obtains the quadratic $(\gamma+1)\,v_{2x}^2-2\gamma\,\frac{P}{m}\,v_{2x}+2(\gamma-1)\left(E-\frac{v_{2y}^2}{2}\right)=0$. For $\mathbf{F}=0$ one root is $v_{2x}=v_{1x}$; the root continuous with it is the physical one (the other is the supersonic-subsonic conjugate state, as in a normal shock). Numerical check: $p_1=101325$ Pa, $\rho_1=1.2$ kg/m^3, $v_{1x}=100$ m/s, $L=0.2$ m, $F_x=30$ N/m, $F_y=-80$ N/m, $\gamma=1.4$ give $v_{2x}=100.16$ m/s, $v_{2y}=3.3333$ m/s, $\rho_2=1.19808$ kg/m^3, $p_2=101155.8$ Pa, with residuals of (E1) to (E4) below 10^{-9} in SI units; the other root gives $p_2=-6843.3$ Pa <0 and is discarded.

PROBLEM CL5

Turbojet on a Testbench: Exit State

Original statement · CL5.pdf

Moving boundaries | Mass, momentum, energy | Heat and work | Thrust

OFFICIAL PROBLEM STATEMENT

The entrance and exit sections of the turboreactor sketched in the figure of CL5.pdf have areas A_1 and A_2. Besides the outer fixed frame, the machine has mobile parts in its interior that extract an amount of work W per unit time from the fluid stream. The combined effects of chemical reaction in the combustion chamber and heat transfer through the walls give a heat transfer per unit time Q to the fluid stream. The axial force on the turboreactor measured on the testbench is $\mathbf{F} = -F\,\mathbf{e}_x$. For given values of W, Q and F and assuming that at the entrance and exit sections the flow properties are uniform, obtain the values of the density ρ_2, pressure p_2 and velocity v_2 in terms of their values p_1, ρ_1 and v_1 at the entrance. (Handwritten addition: no gravity.)

Source: CL5.pdf (page 1). Notes.pdf, Chapter 3: integral mass balance Eq. 3.10, integral momentum balance Eq. 3.35, integral total energy balance Eq. 3.46, force on a body Eq. 3.32.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Degrees of Freedom and Data

- The gas is a perfect gas with specific heat ratio γ ($h = \frac{\gamma}{\gamma-1}\frac{p}{\rho}$), as used in the official solution. Flow properties are uniform over the entrance and exit sections; viscous stresses and conduction are negligible there. No gravity (handwritten addition).
- The flow is steady on average: the rotating parts make it periodic in time at fixed points inside the engine, but through the sections it is steady, so the net accumulation inside the control volume is zero.
- The fuel mass flow is not given and is neglected, so the mass flow at the exit equals that at the entrance (as in the official solution). The ambient pressure p_a acts on the outer surface of the engine and is part of the measured force; p_a is therefore needed as data in addition to those listed.

- The scan prints the direction of the force as $-F\,\mathbf{e}_x$ with $F > 0$ (thrust acts against the exhaust direction). Three unknowns, ρ_2, p_2, v_2, and three equations (mass, axial momentum, energy).

Symbol	Meaning	SI unit
A_1, A_2	entrance and exit areas	m^2
p_1, ρ_1, v_1	entrance state (data)	Pa, $\mathrm{kg/m}^3$, m/s
p_2, ρ_2, v_2	exit state (unknowns)	Pa, $\mathrm{kg/m}^3$, m/s
F	magnitude of the thrust measured on the bench	N
W	power extracted from the fluid by the mobile parts	W
Q	heat per unit time added to the fluid	W
$\dot{m} = \rho_1 v_1 A_1$	mass flow	kg/s

PHASE 2

Coordinates, Frames and Changes of Variable

- Inertial frame fixed to the testbench, with the engine at rest and the axis $\mathbf{e}_x$ along its axis in the direction of the flow.
- Control volume: the gas inside the engine, with boundary Σ_1 (entrance, $\mathbf{n} = -\mathbf{e}_x$, area A_1), Σ_2 (exit, $\mathbf{n} = +\mathbf{e}_x$, area A_2), Σ_3 (inner surface of the fixed casing) and Σ_4 (surface of the mobile parts, moving with velocity $\mathbf{v}_w$). The boundary Σ_4 moves with the blades, so $\mathbf{v}_c = \mathbf{v}_w$ on it. Since the fluid sticks to the walls, $\mathbf{v} = \mathbf{v}_w$ there and $(\mathbf{v} - \mathbf{v}_c)\cdot\mathbf{n} = 0$ on $\Sigma_3 \cup \Sigma_4$. The universal balances (Eqs. 3.10, 3.35, 3.46) are used.
- Since $\int_{\Sigma_c} p_a\,\mathbf{n}\,d\sigma = 0$ on a closed surface, p is replaced by $p - p_a$ in the momentum balance. The measured force $\mathbf{F}$ is the net force of the gas and the atmosphere on the engine, so the force of the engine on the gas is $-\mathbf{F} = +F\,\mathbf{e}_x$ (gauge form, thin casing).
- The power taken from the gas by the mobile parts is defined with the normal $\mathbf{n}$ pointing out of the solid (into the gas), as in CL5.pdf: $W = -\int_{\Sigma_4} p\,\mathbf{v}_w\cdot\mathbf{n}\,d\sigma + \int_{\Sigma_4} \mathbf{v}_w\cdot\bar{\bar{\tau}}'\cdot\mathbf{n}\,d\sigma$, the power of the traction of the gas on the moving solid.

PHASE 3

Step-by-Step Derivation

3.1 Mass

Why this tool: Notes Eq. 3.10 with zero net accumulation. On Σ_3 and Σ_4, $(\mathbf{v}-\mathbf{v}_c)\cdot\mathbf{n}=0$:

$$-\rho_1 v_1 A_1 + \rho_2 v_2 A_2 = 0 \implies \boxed{\rho_2\, v_2\, A_2 = \rho_1\, v_1\, A_1 = \dot{m}}. \qquad \text{(M)}$$

3.2 Axial momentum

Why this tool: the thrust is the force on the engine, which enters the balance of Notes Eq. 3.35 through the unknown wall term. Left-hand side, using (M):

$$\int_{\Sigma_c} \rho\, v_x\, [(\mathbf{v}-\mathbf{v}_c)\cdot\mathbf{n}]\, d\sigma = -\rho_1 v_1^2 A_1 + \rho_2 v_2^2 A_2 = \dot{m}\,(v_2 - v_1).$$

Right-hand side: Σ_1 gives $+(p_1-p_a)A_1$ and Σ_2 gives $-(p_2-p_a)A_2$; viscous stresses are negligible there; the walls give $-\mathbf{F}\cdot\mathbf{e}_x = +F$. Hence

$$\dot{m}\,(v_2-v_1) = (p_1-p_a)A_1 - (p_2-p_a)A_2 + F \implies \boxed{F = \rho_1 v_1 A_1\,(v_2-v_1) - (p_1-p_a)A_1 + (p_2-p_a)A_2} \quad \text{(P)}$$

3.3 Total energy

Why this tool: p_2 and ρ_2 are linked by the heat and work exchanged, which only the energy balance (Notes Eq. 3.46) contains. On Σ_3 the casing is at rest, so no work is done; on Σ_4 the terms $-\int p\,\mathbf{v}\cdot\mathbf{n}\,d\sigma + \int \mathbf{v}\cdot\bar{\bar{\tau}}'\cdot\mathbf{n}\,d\sigma$ of Eq. 3.46 (normal out of the gas) are the power done on the gas by the blades, which is $-W$ by the definition of W in Phase 2; the heat through the walls plus the heat release in the volume is Q. On Σ_1 and Σ_2 the pressure work $p\,\mathbf{v}\cdot\mathbf{n}$ joins the convective flux and turns e into h:

$$\rho_1 v_1 A_1 \left[\left(e_2 + \frac{v_2^2}{2}\right) - \left(e_1 + \frac{v_1^2}{2}\right)\right] = p_1 v_1 A_1 - p_2 v_2 A_2 - W + Q,$$

$$\boxed{\dot{m}\left[\left(\frac{\gamma}{\gamma-1}\frac{p_2}{\rho_2}+\frac{v_2^2}{2}\right)-\left(\frac{\gamma}{\gamma-1}\frac{p_1}{\rho_1}+\frac{v_1^2}{2}\right)\right]=Q-W} \quad \text{(E)}$$

Equations (M), (P) and (E) determine ρ_2, p_2 and v_2.

PHASE 4

Interpretation, Limits and Dimensional Check

- Equation (P) is the classical thrust balance: the thrust is the change of momentum flux plus the pressure imbalance on the two sections. For $p_1 = p_2 = p_a$ it reduces to $F = \dot{m}\,(v_2 - v_1)$.
- Equation (E) says that the stagnation enthalpy of the stream rises by the heat added per unit mass flow and falls by the work extracted per unit mass flow: $Q - W$ is the net energy input.
- Dimensions (SI): $[\dot{m}\,(v_2 - v_1)] = (\mathrm{kg/s})(\mathrm{m/s}) = \mathrm{N}$; $[(Q-W)/\dot{m}] = \mathrm{W}/(\mathrm{kg/s}) = \mathrm{J/kg} = \mathrm{m^2/s^2} = [h] = [v^2]$.
- Numerical illustration (hypothetical data, not a real engine): $A_1 = 0.5$ m^2, $A_2 = 0.3$ m^2, $v_1 = 150$ m/s, $p_1 = 0.9 \times 10^5$ Pa, $\rho_1 = 1.1$ kg/m^3 ($M_1 = 0.443$), $p_a = 10^5$ Pa, $F = 30$ kN, $W = 2$ MW, $Q = 30$ MW, $\gamma = 1.4$ give $\dot{m} = 82.5$ kg/s and the two roots of the extension below: $v_2 = 355.515$ m/s, $p_2 = 126817$ Pa, $\rho_2 = 0.77353$ kg/m^3 ($M_2 = 0.742$) and $v_2 = 597.263$ m/s, $p_2 = 60336$ Pa, $\rho_2 = 0.46043$ kg/m^3 ($M_2 = 1.39$); the residuals of (M), (P) and (E) are below 10^{-6} in SI units.
- Agreement with the official solution: CL5.pdf obtains (M) as $\rho_1 v_1 A_1 = \rho_2 v_2 A_2$, (P) as $F = \rho_1 v_1 A_1 (v_2 - v_1) - (p_1 - p_a)A_1 + (p_2 - p_a)A_2$ and (E) as above, the same equations (reading of the handwriting).

Extension (beyond the official solution): explicit solution

With $K \equiv p_a A_2 + F + (p_1 - p_a) A_1 + \dot{m}\, v_1$, (P) gives $p_2 A_2 = K - \dot{m}\, v_2$ and, with (M), $p_2/\rho_2 = (K - \dot{m}\, v_2)\, v_2/\dot{m}$. Substituting into (E) and using $\frac{1}{2} - \frac{\gamma}{\gamma-1} = -\frac{\gamma+1}{2(\gamma-1)}$ gives the quadratic $(\gamma+1)\, v_2^2 - 2\gamma\, \frac{K}{\dot{m}}\, v_2 + 2(\gamma-1)\, H_0 = 0$ with $H_0 \equiv \frac{\gamma}{\gamma-1}\frac{p_1}{\rho_1} + \frac{v_1^2}{2} + \frac{Q-W}{\dot{m}}$. It has two roots, a subsonic and a supersonic exit state; for a subsonic entrance and heat addition the physical exit state is the subsonic root (the smaller one), since reaching the supersonic root would need a passage through the sonic condition inside the engine, which this overall balance does not describe. For $F = 0$, $Q = W = 0$ and $A_1 = A_2$ one root is $v_2 = v_1$ and the product of the roots is $2(\gamma-1)H_0/(\gamma+1)$, the Prandtl relation of a normal shock.